PPO Workshop

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What is PPO?

1. PPO is REINFORCE with 'clipping'
2. We will only think in terms of *inputs* and *outputs*

Preliminaries

$\pi_\theta(a|s)$ - policy parameterized by θ

$V^{\pi_\theta}(s)$ - value function of π_θ

$L(\theta)$ - loss function to be optimized

Preliminaries - Inputs and Outputs

$\pi_\theta(a|s)$ - make decisions about what action to take given the state it is in and the parameters θ

$V^{\pi_\theta}(s)$ - estimate of future discounted rewards given the state it is in and the policy π_θ

$L(\theta)$ - a number that we want to minimize by changing θ, the lower the number the better the model is doing

As easy as 1, 2, 3

Algorithm 1 PPO, Actor-Critic Style

for iteration$=1, 2, \ldots$ **do**
  for actor$=1, 2, \ldots, N$ **do**
    Run policy $\pi_{\theta_{\text{old}}}$ in environment for T timesteps **(1)**
    Compute advantage estimates $\hat{A}_1, \ldots, \hat{A}_T$ **(2)**
  end for
  Optimize surrogate L wrt θ, with K epochs and minibatch size $M \leq NT$ **(3)**
  $\theta_{\text{old}} \leftarrow \theta$
end for

As easy as 1, 2, 3

Run policy $\pi_{\theta_{\text{old}}}$ in environment for T timesteps

As easy as 1, 2, 3

$$\text{Run policy } \pi_{\theta_{\text{old}}} \text{ in environment for } T \text{ timesteps}$$

1. What are the inputs?
2. What are the outputs?

As easy as 1, 2, 3

$$\text{Run policy } \pi_{\theta_{\text{old}}} \text{ in environment for } T \text{ timesteps}$$

1. What are the inputs? - π_θ, $\pi_{\theta_{old}}$
2. What are the outputs? - $(s_t, a_t, r_t, s_{t+1})_{t=0}^{T} \sim \pi_{\theta_{old}}$

As easy as 1, 2, 3

Compute advantage estimates $\hat{A}_1, \ldots, \hat{A}_T$

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$$\hat{A}_t = -V(s_t) + r_t + \gamma r_{t+1} + \cdots + \gamma^{T-t+1} r_{T-1} + \gamma^{T-t} V(s_T)$$

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1. What are the inputs? - $(s_t, a_t, r_t, s_{t+1})_{t=0}^{T} \sim \pi_{\theta_{old}}$, $V^{\pi_{\theta_{old}}}(s_t)$
2. What are the outputs? - $(\hat{A}_t)_{t=0}^{T}$

As easy as 1, 2, 3

Optimize surrogate L wrt θ, with K epochs and minibatch size $M \leq NT$

No clipping or penalty:	$L_t(\theta) = r_t(\theta)\hat{A}_t$
Clipping:	$L_t(\theta) = \min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta)), 1-\epsilon, 1+\epsilon)\hat{A}_t$
KL penalty (fixed or adaptive)	$L_t(\theta) = r_t(\theta)\hat{A}_t - \beta\,\text{KL}[\pi_{\theta_{\text{old}}}, \pi_\theta]$

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1. What are the inputs? - $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$, $\hat{A}_t$, ϵ
2. What are the outputs? - L_t!!!!!

Pytorch

1. Now we have a loss function L_t that we want to minimize by changing model parameters θ

2. How do we do that?

Pytorch

1. Now we have a loss function L_t that we want to minimize by changing model parameters θ

2. How do we do that? - Gradient descent!

Simple Linear Regression Example

```python
class Lin_Reg_Torch(nn.Module):
    def __init__(self, xdims, ydims) -> None:
        super(Lin_Reg_Torch, self).__init__()
        self.linear = nn.Linear(in_features=xdims, out_features=ydims)
        # pyright: ignore[reportPrivateImportUsage]
        self.optimizer = torch.optim.SGD(params=self.parameters(), lr=1e-5)

    def forward(self, x):
        return self.linear(x)

    def loss(self, y_pred, y):
        return ((y_pred - y) ** 2).mean()
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    def fit(self, X, Y, epochs):
        losses: list[Unknown] = []
        for epoch in range(epochs):
            y_pred: Unknown = self.forward(x=X)
            l: Unknown = self.loss(y_pred, y=Y)
            losses.append(l.item())
            self.optimizer.zero_grad()
            l.backward()
            self.optimizer.step()
            if epoch % 10 == 0:
                print(f'epoch {epoch}: loss = {l}')
        return losses
```

PPO Pytorch

1. PPO Assumes an Actor-Critic architecture

2. One network for the policy π_θ and one for the value function V^{π_θ}

PPO Pytorch

To the code!

Thank you!

Helpful Links

1. Up to data PPO paper - https://arxiv.org/abs/1707.06347
2. Code - https://github.com/GersiD/ML/tree/main/RL/PPO
3. Slides - Email me Gersi.Doko@unh.edu